Quiz1

Graded

Student

Atharva Gupta

Total Points

17 / 20 pts

Question 1

(no title) **4** / 6 pts

1.1 (no title) **2** / 2 pts

- ✔ **+ 1 pt** Correct answer
- ✔ **+ 1 pt** Correct explanation
- **+ 0 pts** Wrong / missing answer (explanation doesn't matter)

1.2 (no title) **2** / 2 pts

- ✔ **+ 1 pt** Correct answer
- ✔ **+ 1 pt** Correct explanation
- **+ 0 pts** Wrong / missing answer (explanation doesn't matter)

1.3 (no title) **0** / 2 pts

- **+ 1 pt** Correct answer
- **+ 1 pt** Correct explanation
- ✔ **+ 0 pts** Wrong / missing answer (explanation doesn't matter)

Question 2

(no title) **6** / 6 pts

2.1 (no title) **3** / 3 pts

- ✔ **+ 1 pt** First call to who in B
- ✔ **+ 1 pt** Second call to who in C
- ✔ **+ 1 pt** Third call to who in A and correct output
- **+ 0 pts** Incorrect / Not attempted

2.2 (no title) **3** / 3 pts

- ✔ **+ 0.5 pts** x.step.step is an instance of B
- ✔ **+ 0.5 pts** x.step is an instance of C
- ✔ **+ 1 pt** x.step.step.who returns "B" + super().who() where super() here is A
- ✔ **+ 1 pt** Correct output
- **+ 0 pts** Incorrect / Not Attempted

Question 3

(no title) **7** / 8 pts

3.1 **(no title)** **2** / 3 pts

+ 3 pts Correct answer

✔ **+ 2 pts** Correct answer with minor errors (explanation from graders required)

+ 1 pt Wrong answer but minor progress

+ 0 pts Wrong / missing answer

you need to prove this implication like via proof by contradiction showing that if rmid were in set L, it would break the if-condition.in case 1 the bound is strictly m < rmid not m <= rmid.

3.2 **(no title)** **1** / 1 pt

✔ **+ 1 pt** Correct

+ 0 pts Wrong / missing answer

write legibly

3.3 **(no title)** **1** / 1 pt

✔ **+ 1 pt** Correct

+ 0 pts Wrong / missing answer

3.4 **(no title)** **3** / 3 pts

✔ **+ 1 pt** Correct recurrence

✔ **+ 1 pt** Correct expression after k unrollings of recurrence

✔ **+ 1 pt** Correct answer in \Theta() notation

+ 0 pts Wrong / missing answer

Name: ATHARVA GUPTA **1** Entry number: 2025MT11539

COL1101-Diwali2026 Quiz 1 (19 Aug 2026) Duration: 1 hour

1. An empty stack and a queue containing elements $0, 1, \ldots, n-1$, in this order from front to back, are given to you (which means that the first dequeue on the queue returns 0 and leaves $1, \ldots, n-1$ in the queue). You are allowed to perform the following operations.

   - *Shift*: dequeue an element from the queue and push it on the stack.
   - *Pop-and-print*: pop an element from the stack and print it. This operation is legal if and only if the stack is non-empty.

   Observe that after exactly n shift and n (legal) pop-and-print operations, the stack and the queue are both empty, and a permutation of $\{0, \ldots n-1\}$ is printed. We say that a permutation $a_0, \ldots, a_{n-1}$ is *producible* if there is a way of interleaving n shift and n pop-and-print operations that makes all the pop-and-print operations legal, and prints the permutation $a_0, \ldots, a_{n-1}$. For example, for $n = 3$, all permutations of $\{0, 1, 2\}$ except $[2, 0, 1]$ are producible.

   Let $n = 6$. Determine whether each of the following permutations of $\{0, 1, 2, 3, 4, 5\}$ is producible, and justify your answer (answers with incorrect justification get 0 points).

   (a) (2 points) $[1, 4, 3, 5, 2, 0]$

   ~~shift till 1~~ : Let stack be S and queue be Q.
   shift till 1 is shifted : S = [0, 1], Q = [2, 3, 4, 5], Pop 1. out = 1
   shift till 4 : S = [0, 2, 3, 4], Q = [5]. Pop 4. out = 4
   ~~shift till~~ Pop 3 : out ← 3, shift till 5 : S = [0, 2, 5]
   Pop all. out ← 5, out ← 2, out ← 0
   output = 1, 4, 3, 5, 2, 0

   (b) (2 points) $[2, 1, 5, 3, 4, 0]$

   this is not possible as after we shift till 5 the number 3, and 4 are still went printed
   ⇒ 3, 4 are in stack ⇒ 4 will be printed before 3 as LIFO, but here 3 is printed first.

   (c) (2 points) $[2, 3, 1, 4, 0, 5]$

   ~~Pop~~ shift till 2 : S = [0, 1, 2], Q = [3, 4, 5]
   Pop 2 : out ← 2, shift 3 : S = [0, 1, 3], Q = [4, 5]
   Pop 3 : out ← 3, S = [0, 1], ~~shift~~ Pop 1 : out ← 1, S = [0]
   shift till 5 : S = [0, 4, 5], Q = [], Pop all ⇒ out ← 5, out ← 4, out ← 0
   final output : 2, 3, 1, 4, 0, 5

Name: _____ **2** Entry number: 2025MT11539

COL1101-Diwali2026 Quiz 1 (19 Aug 2026) Duration: 1 hour

2. Study the following program **carefully**, then answer parts (a)–(c). Assume it runs top to bottom with no errors.

```
class A:
    def who(self):
        return "A"

    def tag(self):
        return "a"


class B(A):
    step = A()

    def who(self):
        return "B" + super().who()

    def tag(self):
        return "b"


class C(A):
    step = B()

    def who(self):
        return "C" + super().who()


class D(B, C):
    step = C()


x = D()
```

(a) (3 points) Give the exact output of `print(x.who())`. List **every** `who` method that runs and the class it lives in, in the order they execute.

print(x.who()) outputs "BCA" because the MRO of D class is in order D→B→C→A and the super func steps forward in the MRO.

I st: B.who, II: C.who, III: A.who

(b) (3 points) Give the exact output of `print(x.step.step.who())`. List the type of the object after each `.step`.

"BA" will be the output of the print.
x is a instance of D.
x.step is a instance of C
x.step.step is a instance of B.
who gets executed of this instance of B.
MRO of this object is (B, A). Therefore, super() is the class A.
⇒ output will be "BA"

3. Recall the *bottom of the valley* problem you solved in the week 2 practical session of this course. A *valley* is a non-empty list v of n numbers for which there exists an index m such that

$$v[0] > \cdots > v[m-1] > v[m] < v[m+1] < \cdots < v[n-1].$$

Given a valley v, the goal was to find the index m at which the minimum of the valley is located.

Here is another python function that finds the bottom of a valley.

```
def bottom_of_the_valley(v):
    left, right = 0, len(v)
    while right - left > 2:
        lmid, rmid = (2*left + right)//3, (left + 2*right)//3
        if v[lmid] < v[rmid]:
            right = rmid
        else:
            left = lmid
    if v[left] < v[right-1]:
        return left
    else:
        return right-1
```

We are going to prove the correctness of the above function by first proving that the statement

`left <= m and m < right`

is a loop invariant for the `while` loop. Then we will analyze the running time of the function. Clearly, the statement is true after 0 iterations of the while loop.

Name: _____ 4 Entry number: _____

COL1101-Diwali2026 Quiz 1 (19 Aug 2026) Duration: 1 hour

(a) (3 points) Suppose that for an arbitrary positive integer i, the statement holds after $i-1$ iterations of the `while` loop. Prove (ideally, in at most four sentences) that the statement holds after i iterations of the `while` loop.

Define two sets L, R. L = [0, m-1], R = [m+1, n-1]
if v[i] > v[i+1] ⇒ i ∈ L | if v[i] < v[i+1] ⇒ i ∈ R
⇒ ∀ j < i, j ∈ L to question. | ⇒ ∀ i s.t. i ≤ j ≤ n-1, j ∈ R
s.t. j < i
case 1: v[mid] < v[mid+1] ⇒ ∀ j s.t. (mid ≤ j) j ∈ R ⇒ m ≤ mid
case 2: v[mid] > v[mid+1] ⇒ ∀ i s.t. (i < mid) i ∈ L ⇒ m > mid
m does not belong in either L or R at every other index. Hence, m = mid.
Therefore, After updating left & right for each case: left ≤ m < right

(b) (1 point) Hence, prove (ideally, in at most two sentences) that if the call to `bottom_of_the_valley` terminates, then it correctly returns `m`, the index at which the valley `v` has its minimum. (For this part, you may assume that `left <= m and m < right` is a loop invariant even if you couldn't prove it in the earlier part.)

m ∈ [left, right-1]
case 1: right - left = 2 : right-1 & left are the only two indices.
case 2: right - left = 1 : Only checks both and returns the valley min.
case 3: right - left < 0 : Not possible as m exist.

(c) (1 point) Why does a call to the function `bottom_of_the_valley` necessarily terminate?

Every iteration, let size = right - left, ∅
if not terminated ⇒ right - left ≥ 3 ⇒ mid > left and mid < right
So, after an iteration, size decreases by atleast 1. Since finite ⇒ will eventually size ≤ 2 ⇒ terminate

(d) (3 points) Let $T(n)$ denote the worst case running time of the function `bottom_of_the_valley` on an input valley `v` with `len(v)` being n. Write down a recurrence for $T(n)$, solve it, and express $T(n)$ using the $\Theta(\cdot)$ notation.

$T(n) = C + T\left(n - \left\lceil \frac{n}{3} \right\rceil\right)$ — comparisons and assignment
$= \lceil \log_{(3/2)}(n) \rceil$ iterations $:= m$
Iteration on reduced segment.
since mid and mid+1 take n.n. of order 3.
$T(n) \simeq C + T\left(\frac{2n}{3}\right) = C_1 + C_2 + T\left(\frac{4n}{9}\right)$
$T(n) \simeq C_1 + C_2 \ldots C_m = \Theta(m)$ → which is $\Theta(\log_{3/2} n)$
Approximate time estimated till termination $\Theta(\log n)$